

A05 – Analyzing Arrival Through the Lens of NLP

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Introduction and Plot Context

Arrival (2016) centers on twelve extraterrestrial spacecraft that appear at locations around the world. In the United States, linguist Louise Banks and physicist Ian Donnelly are recruited to determine why the visitors—called heptapods—have come to Earth. The film is especially relevant to Natural Language Processing (NLP) because Louise must learn a language with no dictionary, bilingual corpus, previous translation, or shared cultural framework. The problem is therefore larger than translation: the team must discover how meaning is represented, establish shared reference points, test interpretations, and communicate under extreme uncertainty (Villeneuve, 2016).

As the group observed, the film repeatedly demonstrates that successful communication depends on more than recognizing symbols. Context, observation, culture, intent, and feedback all matter. These same issues remain central to NLP, where a system can identify patterns in text or symbols without necessarily understanding what a speaker means (Jurafsky & Martin, 2023).

NLP Challenges Depicted in the Film

The clearest NLP challenge is semantic ambiguity. When the heptapods produce a message interpreted as “offer weapon,” military leaders and several governments treat the translation as evidence of a threat. Louise argues that the translated concept may instead mean “tool,” “means,” or something else that does not carry the same hostile intent. This scene illustrates word-sense disambiguation: selecting the intended meaning of a word or symbol from several plausible meanings. A translation can be lexically reasonable while still being pragmatically wrong because the system lacks enough context to infer intent. The consequences in Arrival are extreme, but the underlying challenge is familiar in machine translation, sentiment analysis, and conversational AI.

A second challenge is context. Louise quickly realizes that translating isolated symbols is not enough. Meaning depends on the surrounding message, the physical situation, prior interactions, and the communicative goal. One team contribution connected this directly to ESL experience: understanding the context of a whole sentence often works better than translating one word at a time. Another contribution described a prior sentiment-analysis model that misread Portuguese emotional language, showing how systems can fail when the same apparent word pattern carries different meaning across languages or cultures. These observations parallel a major NLP limitation: models trained on one linguistic or cultural environment may not generalize cleanly to another (Stryker & Holdsworth, 2024).

The film also emphasizes cultural and domain variation. Different countries interact with the heptapods using different assumptions and methods. China’s use of mahjong becomes especially important because Louise worries that a competitive game frames communication around opponents, turns, winning, and losing. Her “hammer and nail” analogy captures the danger: the tool chosen for communication can shape what the user expects to find. In NLP terms, the representation, training data, or classification scheme is not neutral. A system can force unfamiliar input into categories that were built for a different context.

Finally, the film shows the danger of spurious patterns. Ian jokes about possible similarities among landing sites, including low lightning activity and Sheena Easton chart success. The humor makes a real machine-learning point: correlation is not the same as meaningful signal. Both humans and models can discover

regularities that are present in the data but irrelevant to causation or communication. Under pressure, the temptation to treat any pattern as explanatory becomes even stronger.

How Communication Is Built: From Rules to Statistical and Deep NLP

Louise begins with a deliberately simple, rule-based strategy. She writes HUMAN, points to herself, later writes names, and pairs written labels with visible people and actions. When “IAN WALKS” is displayed while Ian physically walks, the team is creating a grounded association between a symbol sequence and an observable event. This resembles a hand-built lexicon or rule-based NLP system in which explicit mappings are established before more complex interpretation is attempted (Manning & Schütze, 1999). It also resembles supervised learning because the team is constructing labeled examples: a known human action is paired with a linguistic representation and then tested through the heptapods’ response.

The film then moves toward statistical pattern analysis. Louise and Ian collect repeated logograms, compare their internal features, examine differences between symbols, and look for consistent associations across interactions. The on-screen visual analysis of points and components resembles feature extraction and corpus analysis: repeated structures are compared to determine which visual elements may carry semantic information. At this stage, the team does not “understand” the language all at once; it accumulates evidence and updates hypotheses as new examples arrive.

Communication is also strongly multimodal. Louise removes her protective suit because she believes the heptapods need to see her. She uses gesture, body position, tapping, pointing, written words, and physical actions to help establish shared attention. Later, when the heptapods respond to the humans’ name-and-gesture routine with distinct symbols of their own, both sides appear to have created a shared convention. This resembles multimodal learning, where meaning is inferred by combining language with visual, spatial, behavioral, or acoustic information rather than relying on text alone (Baltrušaitis et al., 2019).

As the human team gains more data, Louise uses a tablet to assemble heptapod symbols and communicate in their written system. This marks a shift from language understanding to language generation. The device functions like a primitive generative interface: analyzed components are combined into a new output intended to convey meaning back to the heptapods. The progression across the film therefore resembles the historical progression of NLP itself—from manually specified rules, to statistical pattern recognition, to learned representations and generation.

A Language That Does Not Represent Sound

One of the film’s most important linguistic ideas is that heptapod writing is semasiographic: it conveys meaning without directly representing speech sounds. Ian notes that there is no simple correspondence between what the heptapods say and what they write. Their circular logograms appear to encode a complete semantic unit rather than spelling out a spoken sequence. This challenges a human assumption that writing is mainly a visual representation of speech.

The film calls this a nonlinear orthography. Human speech unfolds in time, one sound after another, but the heptapod logogram is produced as a complete circular structure. The writer must seemingly know the whole message before beginning it because the beginning and ending are designed together. This is where the movie moves from language engineering into cognition: if a language is structured nonlinearly, does that structure reflect—or even influence—a nonlinear way of thinking?

This question leads to linguistic relativity, often associated with Benjamin Lee Whorf: the idea that language can influence habitual patterns of thought and perception (Whorf, 1956). *Arrival* dramatizes the idea far beyond ordinary linguistic evidence. As Louise becomes fluent in the heptapod language, she experiences events from her own future as if they were memories. The palindrome Hannah, her daughter’s name, reinforces the film’s

recurring themes of symmetry and beginnings and endings existing as one whole. The film is science fiction, but the underlying question is legitimate: language is not merely a container for pre-existing thoughts; its structures can affect how information is categorized and organized.

Communication, Cooperation, and Interpretation

The statement that the heptapods will help humanity now because they will need humanity's help in 3,000 years reframes the encounter as a non-zero-sum relationship. Earlier, many human decision-makers interpret uncertainty through a threat model. With more semantic context, the same encounter can be understood as cooperation rather than competition. This matters for NLP because interpretation is never separated from the assumptions of the interpreter. A system may assign a label correctly according to its categories while still missing the larger communicative intention.

The international dimension strengthens this point. Different landing sites generate separate observations, translations, and assumptions. When governments stop sharing information, the global "corpus" fragments. In modern data terms, each team has only a partial dataset. Successful interpretation improves when evidence can be compared across contexts, while isolated analysis increases the risk that local assumptions will harden into false certainty.

Reflection: How Arrival Might Look Different in 2026

A particularly important group reflection is how much the technological landscape has changed since *Arrival* was released in 2016. The film predates the transformer architecture introduced in 2017 and the rapid development of today's large language and multimodal models (Vaswani et al., 2017). If the same first-contact scenario occurred today, the human side of the analysis would probably look very different. Researchers could use computer vision to segment logograms, embeddings to represent symbol similarity, clustering to identify recurring structures, multimodal models to connect symbols with gestures and actions, and generative systems to propose candidate translations. If all twelve landing sites shared data, every interaction could contribute to a rapidly growing global corpus.

However, modern AI would not eliminate the central problem in the movie: there is no ground truth. There is no Heptapod-English dictionary and no verified bilingual dataset. A model could discover that two logograms are statistically related without knowing whether the difference means "weapon," "tool," "friend," "enemy," "now," "later," or a concept humans do not possess. Representation learning and embeddings can organize patterns, but statistical proximity is not the same thing as semantic certainty (Mikolov et al., 2013).

Modern generative AI could even introduce a new danger: confident hallucination. A polished system output such as "WEAPON — 83% confidence" might be treated by decision-makers as objective truth even when the underlying evidence supports several plausible interpretations. A responsible system should instead preserve uncertainty, show alternative meanings, explain what evidence supports each possibility, and keep human linguists in the loop. In that sense, the "offer weapon" crisis remains highly relevant to current NLP. Faster pattern recognition does not automatically create understanding.

Conclusion

Arrival presents first contact as a language problem before it becomes a military or scientific one. Louise succeeds because she does not rush from symbols to conclusions. She establishes shared reference points, tests whether meaning has actually been understood, studies patterns, uses multiple modes of communication, and revises interpretations as new evidence appears. The film therefore provides useful parallels to rule-based NLP, statistical NLP, deep representation learning, multimodal systems, and language generation.

The strongest lesson for modern NLP is that communication fails not only when data are missing, but also when people or systems impose assumptions on incomplete data. Context, culture, pragmatics, and intent remain harder than simply matching words or detecting patterns. *Arrival* ultimately asks a deeper question than “Can we translate this language?” It asks what it means to understand another intelligence at all—and whether the structure through which information is represented can change the way an intelligence perceives the world.

Team Collaboration

This report integrates the group’s shared plot summary, scene-based analysis, discussion of ambiguity and cultural variation, reflections on context and language learning, and individual observations about how NLP systems can misread meaning across languages. The combined analysis was organized around the assignment rubric so that the final report addresses NLP challenges, communication methods, NLP approaches and tools, reflection, creative parallels, and current AI implications.

References

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AI Collaboration Notice

Generative AI tools were used to support the organization, synthesis, and refinement of the group’s analysis. AI assistance was used to help consolidate team contributions, improve clarity and structure, connect observations from *Arrival* to NLP concepts, and support editing of the final report. The group reviewed and revised the content for accuracy, relevance, and alignment with the assignment requirements. All interpretations, reflections, and final submitted content remain the responsibility of the student authors.